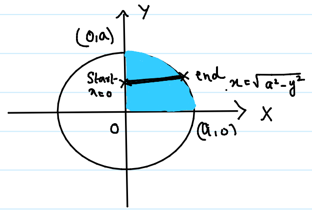
1. Evaluate $\iint xy\,dx\,dy$ over the positive quadrant of the circle $x^2+y^2=a^2$

$\iint xy\,dx\,dy$

$$= \int_{y=0}^{a}\int_{x=0}^{\sqrt{a^2-y^2}} xy\,dx\,dy$$



$$= \int_{y=0}^{a} y\cdot\left(\frac{x^2}{2}\right)_0^{\sqrt{a^2-y^2}} dy$$

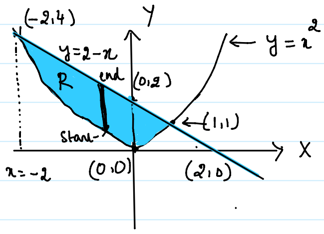
$$= \int_{y=0}^{a} \frac{y}{2}\left\{a^2-y^2\right\} dy = \frac{1}{2}\int_0^a (a^2y-y^3)\,dy$$

$$= \frac{1}{2}\left[a^2\cdot\frac{y^2}{2}-\frac{y^4}{4}\right]_0^a = \frac{1}{2}\left[\frac{a^2}{2}(a^2)-\frac{1}{4}(a^4)\right]$$

$$= \frac{1}{2}\left[\frac{a^4}{2}-\frac{a^4}{4}\right] = \frac{a^4}{8}$$

2. Find the area lying between the parabola $y=x^2$ and the line $x+y=2$

To find point of intersection of $y=x^2 \rightarrow (1)$ and $y=2-x \rightarrow (2)$.



by (1) & (2), $x^2=2-x$

$\Rightarrow x^2+x-2=0$

$\Rightarrow x=\dfrac{-1\pm\sqrt{1+8}}{2} = 1,-2.$

$x=1 \Rightarrow y=1$ and $x=-2 \Rightarrow y=4$ using (1).

Therefore $y=x^2$ and $x+y=2$ intersect at $(1,1)$ and $(-2,4)$

$$\text{Area} = \iint_R dx\,dy = \int_{x=-2}^{2}\int_{y=x^2}^{2-x} dy\,dx$$

$$= \int_{-2}^{2} (y)_{x^2}^{2-x}\,dx$$

$$= \int_{-2}^{2} (2-x-x^2)\,dx$$

$$= \left[2x-\frac{x^2}{2}-\frac{x^3}{3}\right]_{-2}^{2}$$

$$= 2(2+2)-\frac{1}{2}\cancel{(2^2-2^2)}^{0}-\frac{1}{3}(8+8)$$

$$= 8-\frac{16}{3} = \frac{24-16}{3} = \frac{8}{3}$$

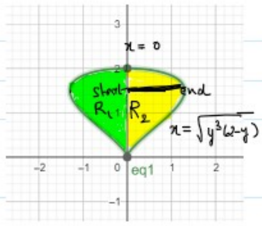
3. By double integration find the whole area of the curve $x^2=y^3(2-y)$.

whole area of the given curve

$= 2 \times$ area in R_1

$= 2 \iint_{R_1} dx\,dy$

$= 2 \int_{y=0}^{2} \int_{x=0}^{\sqrt{y^3(2-y)}} dx\,dy$



$= 2\int_0^2 (x)_0^{y^{3/2}\sqrt{2-y}}\, dy$

$= 2\int_0^2 y^{3/2}(2-y)^{1/2}\, dy$

$= 2\int_0^2 y^{3/2}\left(2\left(1-\frac{y}{2}\right)\right)^{1/2} dy$

$= 2\sqrt{2}\int_0^2 y^{\frac{3}{2}}\left(1-\frac{y}{2}\right)^{1/2} dy$

Let $\frac{y}{2} = t \Rightarrow y = 2t$
$\Rightarrow dy = 2dt$
$y=0 \Rightarrow t=0$; $y=2 \Rightarrow t=1$.

$= 2\sqrt{2}\int_0^1 (2t)^{3/2}(1-t)^{1/2}\, 2\,dt$

$= 4\sqrt{2}\, 2^{3/2}\int_0^1 t^{3/2}(1-t)^{1/2} dt$

$= 4\times 4\int_0^1 t^{\frac{5}{2}-1}(1-t)^{\frac{3}{2}-1} dt$

$= 16\,\beta\left(\frac{5}{2}, \frac{3}{2}\right)$

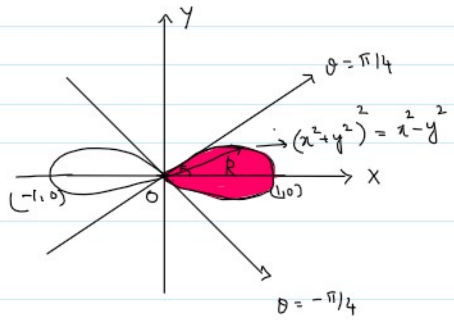
$= \dfrac{16\,\Gamma(\frac{5}{2})\Gamma(\frac{3}{2})}{\Gamma(4)} = \frac{16}{3!} \times \frac{3}{2} \times \frac{1}{2}\sqrt{\pi} \times \frac{1}{2} \times \sqrt{\pi} = \underline{\underline{\pi}}$

4. Evaluate $\iint_R \frac{1}{(x^2+y^2+1)^2}\, dx\,dy$ over one loop of lemniscate $(x^2+y^2)^2 = x^2-y^2$.

Let $x = r\cos\theta$ $y = r\sin\theta$ $dx\,dy = r\,dr\,d\theta$.

In R, r varies from 0 to $\sqrt{\cos 2\theta}$

θ varies from $-\frac{\pi}{4}$ to $\frac{\pi}{4}$



$\frac{1}{(1+x^2+y^2)^2} = \frac{1}{(1+r^2)^2}$

$\therefore \iint_R \frac{1}{(x^2+y^2+1)^2}\, dx\,dy = \int_{\theta=-\frac{\pi}{4}}^{\pi/4}\int_{r=0}^{\sqrt{\cos 2\theta}} \frac{r\,dr\,d\theta}{(1+r^2)^2}$

$(x^2+y^2)^2 = x^2-y^2$
$\Rightarrow r^4 = r^2(\cos^2\theta - \sin^2\theta)$
$\Rightarrow r^2 = \cos 2\theta$

$= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{2}\left(\frac{-1}{1+r^2}\right)_0^{\sqrt{\cos 2\theta}}$

π 1.

$$= -\frac{1}{2}\int_{-\pi/4}^{\pi/4}\left(\frac{1}{1+\cos 2\theta} - 1\right)d\theta$$

$$= -\frac{1}{2}\int_{-\pi/4}^{\pi/4}\frac{-\cos 2\theta}{1+\cos 2\theta}\,d\theta$$

$$= +\frac{1}{2}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{2\cos^2\theta - 1}{\cancel{1}+2\cos^2\theta-\cancel{1}}\,d\theta$$

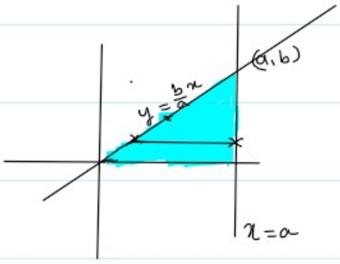
$$= \frac{1}{2}\int_{-\frac{\pi}{4}}^{\pi/4}\left(1-\frac{1}{2}\sec^2\theta\right)d\theta$$

$$= \frac{1}{2}\left(\theta - \frac{1}{2}\tan\theta\right)_{-\frac{\pi}{4}}^{\frac{\pi}{4}} = \frac{1}{2}\left\{\frac{\pi}{2} - \frac{1}{2}(2)\right\}$$

$$= \frac{1}{2}\left\{\frac{\pi}{2} - 1\right\} = \frac{\pi}{4} - \frac{1}{2} \;//$$

5. Evaluate $\int_0^a\int_0^{\frac{bx}{a}} x$

$$\int_0^a\int_0^{\frac{bx}{a}} x\,dy\,dx = \int_{y=0}^{b}\int_{x=\frac{ay}{b}}^{a} x\,dx\,dy$$



$$= \int_0^b \left(\frac{x^2}{2}\right)_{\frac{ay}{b}}^{a} dy$$

$$= \int_0^b \frac{1}{2}\cdot\left(a^2 - \frac{a^2y^2}{b^2}\right)dy = \frac{1}{2}\int_0^b\left(a^2 - \frac{a^2}{b^2}\cdot y^2\right)dy$$

$$= \frac{a^2}{2}\int_0^b\left(1-\frac{y^2}{b^2}\right)dy = \frac{a^2}{2}\left(y - \frac{y^3}{3b^2}\right)_0^b$$

$$= \frac{a^2}{2}\left(b - \frac{b^3}{3b^2}\right) = \frac{a^2}{2}\left(b - \frac{b}{3}\right) = \frac{a^2 b}{3}$$

6. Evaluate $\int_0^1\int_{y^2}^1\int_0^{1-x} x\,dz\,dx\,dy$.

$$\int_0^1\int_{y^2}^1\int_0^{1-x} x\,dz\,dx\,dy = \int_0^1\int_{y^2}^1 x\,(z)_0^{1-x}\,dx\,dy$$

$$= \int_0^1\int_{y^2}^1 x\{1-x\}\,dx\,dy$$

$$= \int_0^1\int_{y^2}^1 (x-x^2)\,dx\,dy$$

$$= \int_0^1\left(\frac{x^2}{2} - \frac{x^3}{3}\right)_{y^2}^1 dy$$

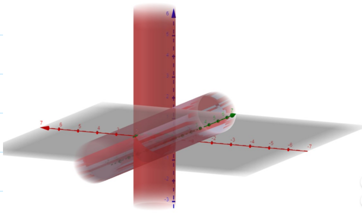
$$= \int_0^1 \left(\frac{1}{2} - \frac{1}{3} - \frac{y^4}{2} + \frac{y^6}{3} \right) dy$$

$$= \int_0^1 \left(\frac{1}{6} - \frac{y^4}{2} + \frac{y^6}{3} \right) dy$$

$$= \frac{1}{6} (y)_0^1 - \frac{1}{2} \cdot \frac{(y^5)_0^1}{5} + \frac{(y^7)_0^1}{3 \times 7}$$

$$= \frac{1}{6} - \frac{1}{10} + \frac{1}{21} = \frac{4}{35} \ //$$

7 Find the volume enclosed by the cyllinders $x^2+y^2 = 2ax$ and $x^2+z^2 = 2ax$.



$$\text{Volume} = 2 \int_{x=0}^{2a} \int_{y=-\sqrt{2ax-x^2}}^{\sqrt{2ax-x^2}} z \, dx dy = 2 \int_0^{2a} \cdot 2 \int_0^{\sqrt{2ax-x^2}} \sqrt{2ax-x^2} \, dy \, dx$$

$$= 4 \int_0^{2a} \sqrt{2ax-x^2} \cdot \left[y \right]_0^{\sqrt{2ax-x^2}} dx = 4 \int_0^{2a} (2ax - x^2) dx$$

$$= 4 \left(\frac{2ax^2}{2} - \frac{x^3}{3} \right)_0^{2a} = 4 \left[a (x^2)_0^{2a} - \frac{1}{3} (x^3)_0^{2a} \right]$$

$$= 4 \left[4a^3 - \frac{8a^3}{3} \right] = \frac{16a^3}{3} \ //$$

8. Show that $\iiint \frac{dx dy dz}{\sqrt{a^2-x^2-y^2-z^2}} = \frac{\pi^2 a^2}{8}$ where region of integration is first octant of the sphere $x^2+y^2+z^2 = a^2$.

$$\iiint_R \frac{dx \, dy \, dz}{\sqrt{a^2-x^2-y^2-z^2}}$$ where R is the first octant of the sphere $x^2+y^2+z^2 = a^2$.

Let us make use of spherical coordinates — $x = r\sin\theta\cos\phi$ $y = r\sin\theta\sin\phi$ & $z = r\cos\theta$.

$\therefore x^2+y^2+z^2 = r^2$.

In R, r varies from 0 to a

θ varies from 0 to $\pi/2$

ϕ varies from 0 to $\pi/2$

$dx dy dz = r^2 \sin\theta \, dr d\theta d\phi$

$$\therefore \iiint_R \frac{dx dy dz}{\sqrt{a^2-x^2-y^2-z^2}} = \int_{\phi=0}^{\pi/2} \int_{\theta=0}^{\pi/2} \int_{r=0}^{a} \frac{r^2 \sin\theta \, dr d\theta d\phi}{\sqrt{a^2-r^2}}$$

$$= \int_{\phi=0}^{\frac{\pi}{2}} d\phi \cdot \int_{\theta=0}^{\frac{\pi}{2}} \sin\theta \, d\theta \cdot \int_{r=0}^{a} \frac{r^2 dr}{\sqrt{a^2-r^2}}.$$

$r = a\sin\theta$

$dr = a\cos\theta \, d\theta$.

$r = a \Rightarrow \theta = \pi/2$

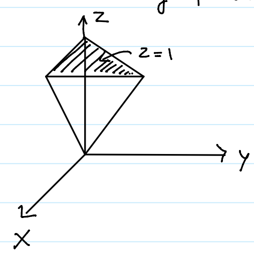
$r = 0 \Rightarrow \theta = 0.$

$$= \frac{\pi}{2} \left(-\cos\theta\right)_0^{\pi/2} \cdot \int_0^{\pi/2} \frac{a^2 \sin^2\theta \cdot a\cancel{\cos\theta}\, d\theta}{\cancel{a\cos\theta}}$$

$$= \frac{\pi}{2}(0+1)\, a^2 \cdot \int_0^{\pi/2} \sin^2\theta\, d\theta$$

$$= \frac{\pi a^2}{2} \cdot \frac{1}{2}\frac{\pi}{2} = \frac{\pi^2 a^2}{8}$$

9. Find the moment of inertia about z axis of a homogeneous tetrahedron bounded by planes $x=0$, $y=0$, $z = x+y$ and $z=1$.



Moment of inertia about z axis $= I_z = \iiint_V \rho(x^2+y^2)\, dx\, dy\, dz.$

Let $\rho = 1$, $\therefore I_z = \displaystyle\int_{x=0}^{1} \int_{y=0}^{1-x} \int_{z=x+y}^{1} (x^2+y^2)\, dz\, dy\, dx$

$$= \int_0^1 \int_0^{1-x} (z)_{x+y}^{1} (x^2+y^2)\, dy\, dx$$

$$= \int_0^1 \int_0^{1-x} (1-x-y)(x^2+y^2)\, dy\, dx$$

$$= \int_0^1 \int_0^{1-x} \left\{ x^2(1-x) - x^2 y + y^2(1-x) - y^3 \right\} dy\, dx$$

$$= \int_0^1 \left\{ x^2(1-x)(y)_0^{1-x} - x^2\left(\frac{y^2}{2}\right)_0^{1-x} + (1-x)\left(\frac{y^3}{3}\right)_0^{1-x} - \left(\frac{y^4}{4}\right)_0^{1-x} \right\} dx$$

$$= \int_0^1 \left\{ x^2(1-x)(1-x) - \frac{x^2}{2}(1-x)^2 + \frac{(1-x)}{3}(1-x)^3 - \frac{1}{4}(1-x)^4 \right\} dx$$

$$= \int_0^1 \left\{ x^2(1-x)^2 - \frac{x^2}{2}(1-x)^2 + \frac{(1-x)^4}{3} - \frac{(1-x)^4}{4} \right\} dx$$

$$= \int_0^1 \left\{ \frac{(x^2-2x+1)x^2}{2} + \frac{(1-x)^4}{12} \right\} dx.$$

$\left.\frac{(x-1)^5}{5}\right|_0$

$$= \left[\frac{1}{2}\left\{ \frac{x^5}{5} - \frac{2x^4}{4} + \frac{x^3}{3} \right\} + \frac{(1-x)^5}{-60} \right]_0^1$$

$$= \frac{1}{2}\left\{ \frac{1}{5} - \frac{1}{2} + \frac{1}{3} \right\} + \frac{1}{60} = \frac{1}{2}\left\{ \frac{6-15+10}{30} \right\} + \frac{1}{60} = \frac{1}{60} + \frac{1}{60} = \frac{1}{30}$$